Automatic Smart Tea Maker

IDE 438

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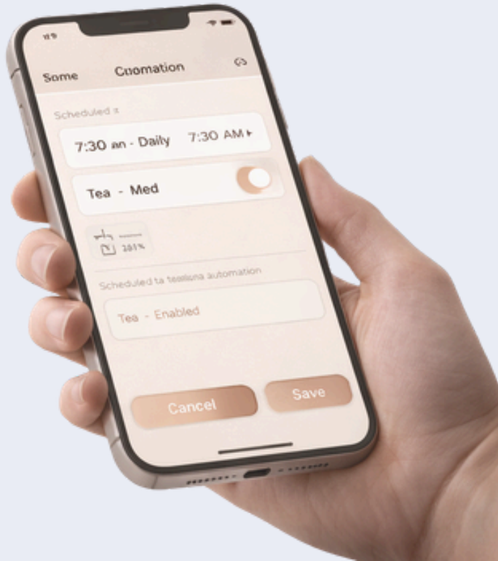

PROJECT OVERVIEW

The project :
Explores how everyday tea preparation can be redesigned to reduce direct interaction with boiling water and steam while preserving the cultural ritual of tea drinking.

Proposes a smart tea maker system that automates parts of the brewing process and allows users to schedule tea preparation in advance.

Domestic safety
Handling boiling water and steam during tea preparation can create safety risks in the home environment. The design focuses on reducing these risks.

Tea ritual
Tea drinking is an important daily routine in many households. The project aims to support this cultural habit while simplifying the preparation process.

Automation
Automating parts of the brewing process helps reduce direct contact with hot water and simplifies everyday use.

Scheduled brewing
The system allows tea to be prepared at a specific time, aligning tea preparation with daily routines.

DESIGN PROCESS TIMELINE

WEEK 01

AI design ideas
Lock mechanisms
Visual sketches

AI role
- generate kettle concepts
- explore child-lock mechanisms
- generate visual alternatives

Designer role
- evaluate usability
- select twist-lock concept

WEEK 02

Early Upgrade

AI role
- explain safe use definition
- analyze interaction risks
- identify dangerous transitions

Designer role
- interpret insights
- connect them to product interaction

WEEK 03

User Feedback

AI role
- synthesize burn injury research
- summarize ergonomic studies
- identify opportunity areas

Designer role
- prioritize insights
- define design tensions

WEEK 04

Advanced Stage

AI role
- evaluate smart tea maker concept
- analyze product examples
- identify engineering risks

Designer role
- refine system idea
- decide design direction

WEEK 05

Future Plan

AI role
- generate kettle concepts
- explore child-lock mechanisms
- generate visual alternatives

Designer role
- evaluate usability
- select twist-lock concept

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DESIGN MILESTONES

01

Problem Fit

Identified whether kettle-related injuries originate from technical failure or everyday interaction patterns.
Research shows that most risks occur during transitional moments such as pouring, lid opening, and steam release.

02

Safety Understanding

Analyzed how thermal risks such as steam exposure and hot water spills occur during daily use.
Understanding these risks helped define safety as an interaction problem rather than a purely technical one.

03

Design Opportunity

Translated research insights into design opportunities such as controlled movement, steam management, and behavior-guiding interaction.

04

Concept Direction

Focused the project on safer transitional interactions around hot water and steam in child-present domestic environments.

05

System Exploration

Explored a smart tea maker system that automates tea preparation and aligns with everyday domestic routines.

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ESSENTIAL PROPMTS

01

How feasible is a smart tea maker with scheduled brewing?
AI Role :EXPANDER

AI Response Summary:

AI evaluated existing smart tea appliances and analyzed feasibility.

Reflection:

The project evolved from a safer kettle into a smart tea preparation system.

Design Decision:

Develop a tea maker with automated brewing and scheduling.
Accepted
Scheduled brewing.
Rejected
Traditional stacked kettle system.
Transformed
Kettle redesign → automated tea system.

04

How can I prototype my tea maker concept without building a full electronic system?
AI Role :CLARIFIER

AI Response Summary:

AI suggested creating a form prototype combined with interaction simulation.

Reflection:

Prototyping should test interaction logic rather than technical engineering.

Design Decision:

Build a simplified prototype demonstrating brewing flow and user interaction.

ESSENTIAL PROPMTS

03

What does safe use actually mean for a tea maker?
AI Role :CLARIFIER

AI Response Summary:

AI explained that kettle safety involves every interaction stage including filling, boiling, pouring, and cooling.

Reflection:

Safety should be embedded in interaction rather than communicated through warnings.

Design Decision:

Focus on behaviour-guided safety.
Accepted
Interaction-based safety.
Rejected
Warning labels.
Transformed
Safety feature → interaction design strategy.

02

What are the main safety risks associated with kettles in domestic environments?
AI Role :REFRAMER

AI Response Summary:

AI synthesized research showing that most accidents occur during transitions such as pouring and lid opening.

Reflection:

Risk is concentrated in movement rather than stable states.

Design Decision:

Focus on transitional interactions.
Accepted
Steam risk as primary issue.
Rejected
Electrical safety as main focus.
Transformed
Burn risk → interaction design opportunity.

AI Collaboration

01

concept exploration

AI helped generate alternative design ideas and safety mechanisms during the early concept development stage.

02

research synthesis

AI supported the process by summarizing scientific literature and identifying key safety insights related to kettle use.

03

interaction analysis

AI helped analyze how users interact with kettles and identify risky moments such as pouring and steam exposure.

04

prototype planning

AI suggested strategies for building a simplified prototype that focuses on demonstrating interaction and system logic.

05

feasibility study

AI evaluated the practicality of the smart tea maker concept by comparing existing products and technical possibilities.

06

Decision filtering (accept / reject / transform)

Critically evaluated AI suggestions by selecting relevant ideas, rejecting unsuitable ones, and transforming others into new design directions.

FINAL INSIGHT

1

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3

4

Core Insight

Research and interaction analysis revealed that most kettle related accidents do not originate from technological failure, but from everyday interaction with hot water and steam during routine activities such as pouring, lifting, and opening the lid.

Design Response

The design focuses on:

- controlled interaction with hot liquids
- reduced direct contact with boiling water and steam
- automated tea preparation through scheduled brewing

Design Implication

This finding suggests that improving kettle safety should not focus only on technical performance, but on how users interact with the product during everyday transitions.

Therefore, safety must be embedded into interaction rather than relying on user attention or warning labels.